Labwork 1b

I. Matlab function

1. Solve, fzero

Exercise 1:

Use either *solve* or *fzero*, as appropriate, to solve the following equations

(a) $67x + 32 = 0$ (exact solution).

(b) $67x + 32 = 0$ (numerical solution to 15 places) – hint: *format long*.

(c) $x^3 + px + q = 0$. (Solve for x in terms of p and q.)

(d) $e^x = 8x - 4$ (all real solutions). It helps to draw a picture first.

2. Plot

Exercise 1

a. Use *plot* to graph the following functions.
(a) $y = x^3 - x$ for $-4 \le x \le 4$.
(b) $y = \tan(x/2)$ for $-\pi \le x \le \pi$, $-10 \le y \le 10$. (Hint: first draw the plot; then use axis.)
(d) $y = e^{-x/2}$ and $y = x^4 - x^2$ for $-1.5 \le x \le 1.5$ (on the same set of axes).

II. Matlab-numerical method

1. Bisection method

Given an M-file to implement the bisection method:

```
function [root,fx,ea,iter]=bisection(func,xl,xu,es,maxit)
% bisect: root location zeroes
% [root,fx,ea,iter]=bisect(func,xl,xu,es,maxit,p1,p2,...):
% uses bisection method to find the root of func
% input:
% func = name of function
% xl, xu = lower and upper guesses
% es = desired relative error (default = 0.0001%)
% maxit = maximum allowable iterations (default = 50)
% p1,p2,... = additional parameters used by func
% output:
% root = real root
% fx = function value at root
% ea = approximate relative error (%)
% iter = number of iterations
if nargin<3,error('at least 3 input arguments required'),end
test = func(xl)*func(xu);
if test>0,error('no sign change'),end
if nargin<4||isempty(es), es=0.0001;end
if nargin<5||isempty(maxit), maxit=50;end
```

```
iter = 0; xr = xl; ea = 100;
while (1)
xrold = xr;
xr = (xl + xu)/2;
iter = iter + 1;
if xr ~= 0,ea = abs((xr - xrold)/xr) * 100;end
test = func(xl)*func(xr);
if test < 0
xu = xr;
elseif test > 0
xl = xr;
else
ea = 0;
end
if ea <= es || iter >= maxit,break,end
end
root = xr;
fx = func(xr);
```

Exercise 1

Use the function file above to find the square root of 2, taking 1 and 2 as initial values of x_l and x_u. Continue bisecting until the maximum error is less than 0.05

Exercise 2

Use the function file above to Determine the positive real root of **$\ln(x^2) = 0.7$** using three iterations of the bisection method, with initial guesses of $x_l = 0.5$ and $x_u = 2$

Exercise 3

Write your script file to implement the Bisection method for solving the equation: $x^2 + 2x - 3 = 0$

2. Newton Raphson method

Newton's method may be used to solve a general equation f(x) = 0 by repeating the assignment

$$\text{x becomes } x - \frac{f(x)}{f'(x)}$$

where *f*'(x) (i.e., d*f*/d*x*) is the first derivative of f(x). The process continues until successive approximations to x are close enough

Suppose that $f(x) = x^3 + x - 3$

that is, we want to solve the equation $x^3 + x - 3 = 0$ (another way of stating the problem is to say we want to find the zero of f(x)). We have to be able to differentiate f(x), which is quite easy here:

f '(x) = $3x^2 + 1$. We could write inline objects for both f(x) and f'(x), but for this example we will use function M-files instead.

Here is a structure plan to implement Newton's method:

1. Input starting value *x*0 and required relative error *e*

2. While relative error $|(x_k - x_{k-1})/x_k| \geq e$, repeat up to, say, $k = 20$:

$$x_{k+1} = x_k - f(x_k)/f'(x_k)$$

Print x_{k+1} and $f(x_{k+1})$

3. Stop

Exercise 1

Use the Editor to create and save (in the current MATLAB directory) the function file **f.m** as follows:

```
function y = f(x)
 y = x^3 + x - 3;
```

Then create and save another function file **df.m**:

```
function y = df(x)
y = 3*x^2 + 1;
```

Now write a separate script file, **newtgen.m** (in the same directory), that will stop either when the relative error in x is less than 10^{-8} or after, say, 20 steps.

Exercise 2

Determine the positive root of $f(x) = x^{10} - 1$ using the Newton Raphson method and an initial guess of $x = 0.5$

Use a simple plot (Matlab) of the first few iterations (helpful in providing insight) to prove that after the first poor prediction, the technique is converging on the true root of 1, but at a very slow rate.

Exercise 3

Write your script file to implement the Newton raphson method for solving the equation: $x^2 + 2x - 3 = 0$.